

Apache Sqoop and Hive Assignments

Commands Reference

1. Grant DB access permission in MySQL

```
GRANT ALL PRIVILEGES ON *.* TO 'msis'@'localhost' IDENTIFIED BY 'mahe';
```

2. List all databases in MySQL through Sqoop

```
sqoop list-databases --connect jdbc:mysql://localhost?useSSL=false --username msis  
--password mahe
```

3. List all tables in selected database

```
sqoop list-tables --connect jdbc:mysql://localhost/retail_db?useSSL=false --username msis  
--password mahe
```

4. Import table into HDFS using target-dir method

```
sqoop import --connect jdbc:mysql://localhost/retail_db?useSSL=false --username msis  
--password mahe --table orders --target-dir '/test_order'
```

5. Import table into HDFS using warehouse-dir method

```
sqoop import --connect jdbc:mysql://localhost/retail_db?useSSL=false --username msis  
--password mahe --table orders --warehouse-dir '/retail_db'
```

6. Import partial data from a specified table

```
sqoop import --connect jdbc:mysql://localhost/retail_db?useSSL=false --username msis  
--password mahe --table orders --target-dir '/partial' --where "order_id>68800"
```

7. Import only selected columns from a table

```
sqoop import --connect jdbc:mysql://localhost/retail_db?useSSL=false --username msis  
--password mahe --columns order_id,order_customer_id --table orders --target-dir  
'/orders12'
```

8. Import all tables from a given database.

```
sqoop import-all-tables --connect jdbc:mysql://localhost/retail_db?useSSL=false  
--username msis --password mahe --warehouse-dir '/retail_db_alltables'
```

9. Exclude some tables while importing all tables

```
sqoop import-all-tables --connect jdbc:mysql://localhost/retail_db?useSSL=false  
--username msis --password mahe --warehouse-dir '/retail_db1' --exclude-tables  
orders,customers
```

10. Export table from HDFS to local database

```
sqoop export --connect jdbc:mysql://localhost/test1?useSSL=false --username msis  
--password mahe --table test_table --export-dir '/user/test1'
```

11. Create Hive table in HDFS

```
hive  
CREATE DATABASE db1;  
USE db1;  
CREATE TABLE departments (department_id INT, department_name VARCHAR(45)) ROW FORMAT  
DELIMITED FIELDS TERMINATED BY ',' STORED AS TEXTFILE;
```

12. Load data from HDFS to Hive table

```
LOAD DATA INPATH '/retail_db_alltables/departments' INTO TABLE db1.departments;
```

13. Run SQL query in Hive

```
SELECT * FROM db1.departments;
```

Other useful Hive SQL examples:

```
SELECT department_id, department_name FROM db1.departments;
```

```
SELECT COUNT(*) FROM db1.departments;
```

Common terminals

MySQL terminal → mysql

Sqoop commands → Hadoop/YARN/Linux terminal

Hive commands → hive